

# **Elementary Knowledge on Ground Water**

## **Module – 2**

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### **Aquifer:**

An aquifer is a geological formation made up of enough permeable material that allows the storage of water and at the same time it will allow the movement of water through it under ordinary conditions. e.g., **Sand, Gravels**

### **Aquiclude:**

An aquiclude may be defined as a geological formation of relatively impermeable material which permits storage of water but it is not capable of transmitting water through it. e.g., **clay**

### **Aquitard:**

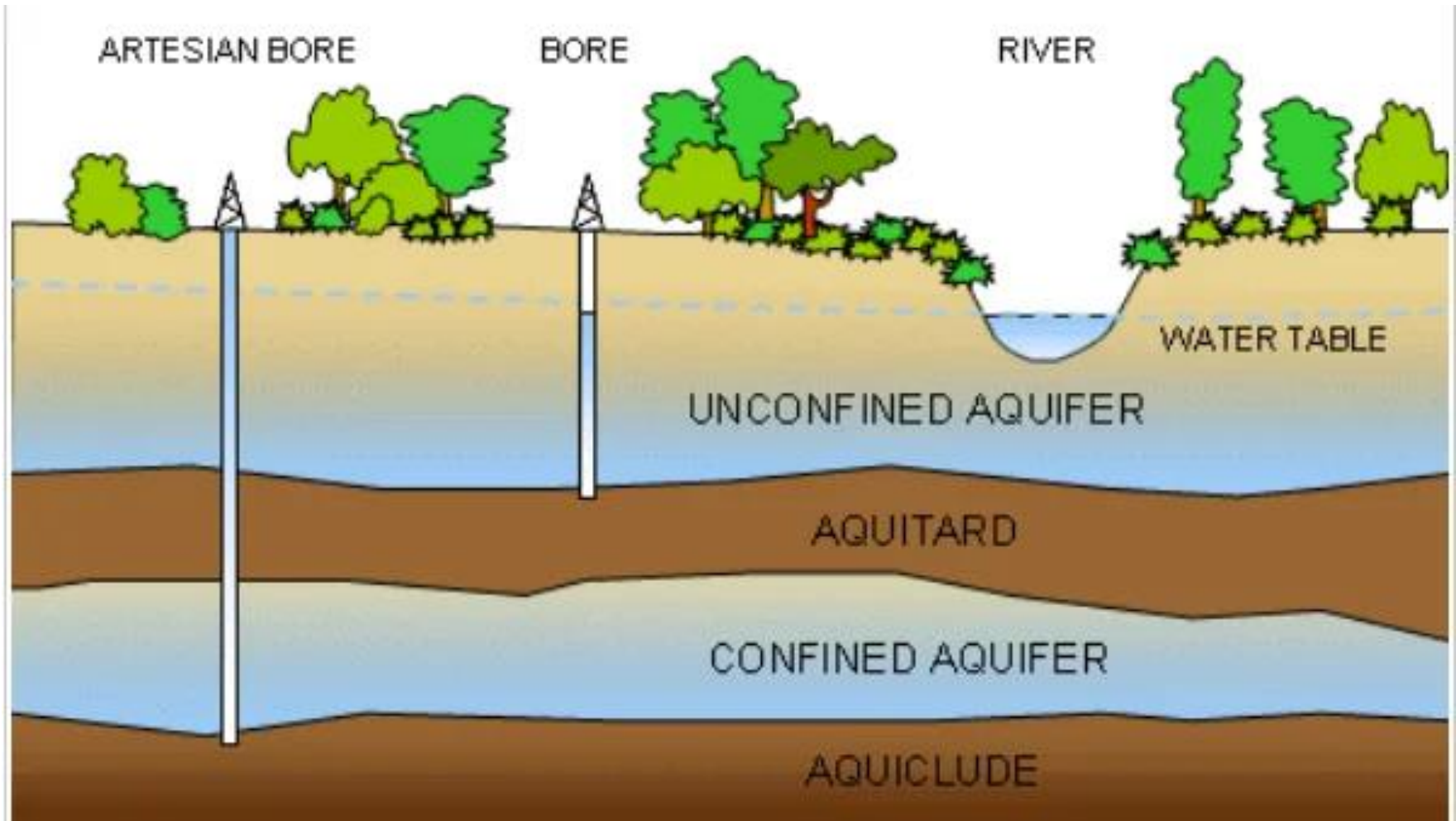
Aquitard is a geological formation of poor permeability, but through which seepage is possible and hence it won't yield water freely to wells. It may transmit vertically appreciable quantities of water to or from adjacent aquifers.

**Sandy clay** is an example of the aquitard.

### **Aquifuge:**

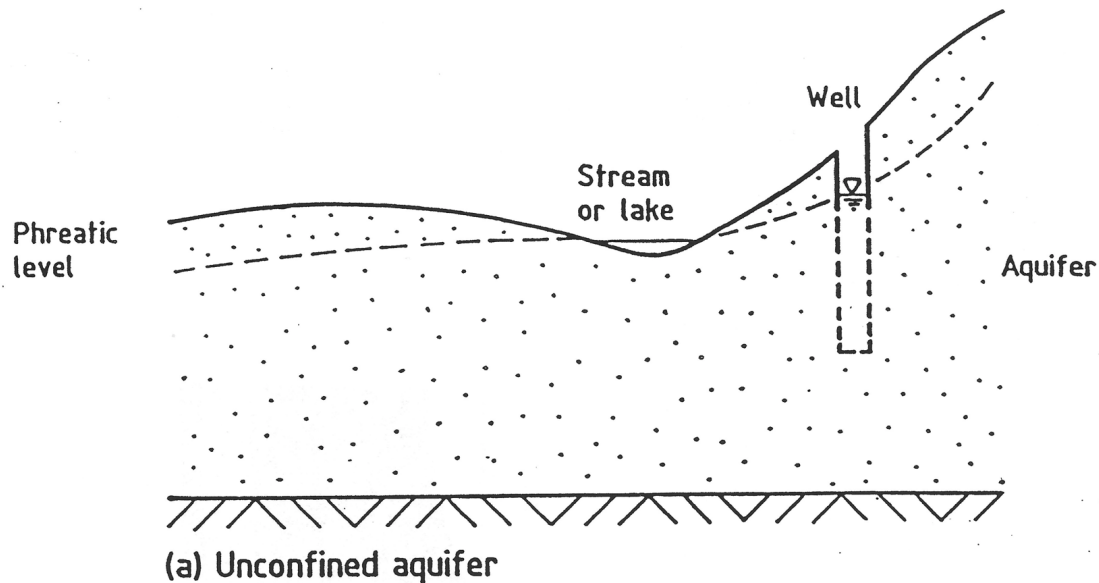
An aquifuge may be defined as a geological formation of impermeable material which neither contains nor transmits water through it. For example **Solid rocks**

**General Aquifers:** An aquifer is an underground layer of porous rocks or permeable rocks that store and retain groundwater levels in the soil.



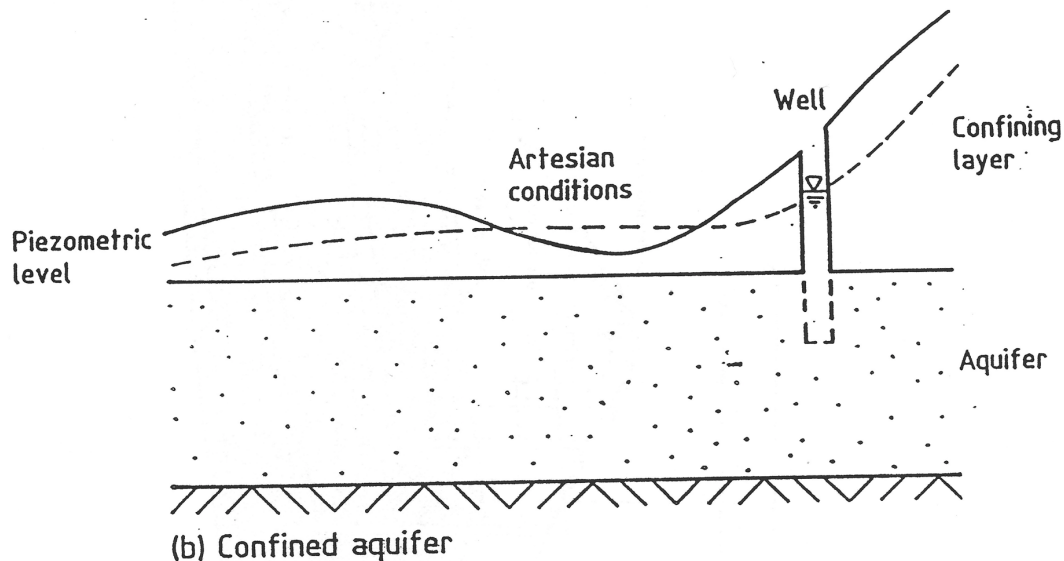
**Unconfined Aquifers:** An unconfined aquifer, also known as a water-table aquifer, is an aquifer with the water table as the upper boundary.

- It is one, in which water is not under any pressure or say water table surface coincides with the upper surface zone of saturation.
- Water seeps directly from the surface into the aquifer.
- Easily recharged by rain and surface water but also more vulnerable to pollution.
- Example: Shallow wells often tap into unconfined aquifers.



**Confined Aquifers:** A confined aquifer is one in which the water table is under pressure greater than atmospheric pressure due to overlying impermeable strata. It is also known as an artesian or pressure aquifer

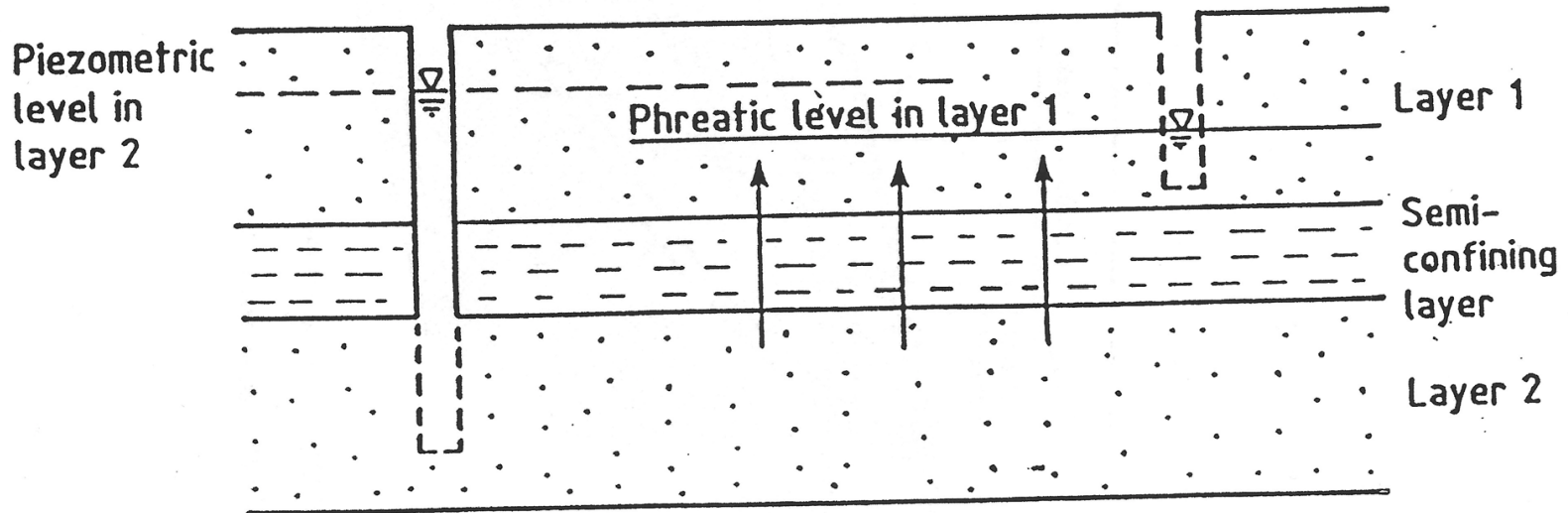
- Sandwiched between two impermeable layers (such as clay or rock).
- Water is under pressure, and when tapped, it may flow naturally without pumping (artesian well).
- Less affected by surface pollution but takes longer to recharge.



An artesian well is one that yields water to the ground surface without pumping. Artesian wells are found in confined aquifers.

**Semi – Confined (Leaky) Aquifers:** If two aquifers are separated by an **aquitard** then they are semi-confined, or part of a leaky aquifer system.

- Semi-confined aquifers are intermediate between confined and unconfined aquifers.
- Less affected by surface pollution but takes longer to recharge.



(c) Leaky aquifer

## Quality of water:

- Water quality refers to the physical, chemical, and biological characteristics of water, determining its suitability for drinking, cooking, and other uses.
- Contaminated water can cause serious health risks, making clean water essential for public health and well-being.

### Physical Characteristics:

**Color:** Clean water should be clear and free from discoloration.

**Odor & Taste:** Pure water is odorless and tasteless; bad smells may indicate contamination.

**Turbidity:** High turbidity means the water has suspended particles, which may carry harmful microbes.

**Temperature:** Warm water can promote bacterial growth, while cold water slows it.

## Chemical Characteristics:

**pH Level:** Safe drinking water should have a pH between **6.5 and 8.5**.

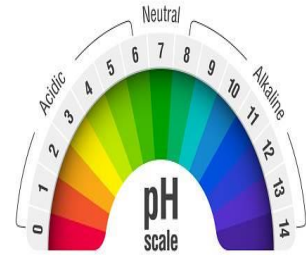
**Dissolved Minerals:** Essential minerals like calcium and magnesium are beneficial, but excessive amounts can cause health problems.

**Toxic Chemicals:** Heavy metals (lead, arsenic, mercury), nitrates, pesticides, and industrial chemicals can be harmful.

**Chlorine & Fluoride:** Used in water treatment; excessive amounts can cause health issues.

## Biological Characteristics:

**Bacteria & Viruses:** Harmful microbes like E. coli, Salmonella, and cholera can cause severe infections.



# Impact of water quality on Human Health

## Short-Term Effects:

**Diarrheal Diseases:** Contaminated water can cause cholera, dysentery, and gastroenteritis.

**Vomiting & Stomach Pain:** Due to microbial contamination or chemical pollutants.

**Skin Infections & Rashes:** Caused by exposure to polluted water.

## Long-Term Effects:

**Kidney & Liver Damage:** Heavy metals like arsenic and lead accumulate in the body.

**Cancer Risk:** Exposure to industrial chemicals and pesticides can increase cancer rates.

**Developmental Issues:** Lead-contaminated water affects brain development in children.

**Weakened Immune System:** Long-term exposure to harmful bacteria and chemicals can make the body more vulnerable to diseases.

# RAIN WATER HARVESTING

## Collecting and storing rainwater for reuse

- The rain water harvesting may be defined as the technique of collection and storage of rain water at surface or in sub-surface aquifer before it is lost as surface run off.
- Rainwater harvesting is the practice of collecting and storing rainwater for reuse.
- It involves collecting rain from a roof-like surface and redirecting it to a tank, well, or other storage system.
- The goal is to conserve water, manage stormwater runoff, and ensure sustainable water resources



## **Advantages of Rain Water Harvesting**

- Promotes adequacy of underground water
- Mitigates the effect of drought
- Reduces soil erosion as surface run-off is reduced
- Decreases load on storm water disposal system
- Reduces flood hazards
- Improves ground water quality / decreases salinity (by dilution)
- Prevents ingress of sea water in subsurface aquifers in coastal areas
- Improves ground water table, thus saving energy (to lift water)
- The cost of recharging subsurface aquifer is lower than surface reservoirs
- The subsurface aquifer also serves as storage and distribution system
- No land is wasted for storage purpose and no population displacement is involved
- Storing water underground is environment friendly

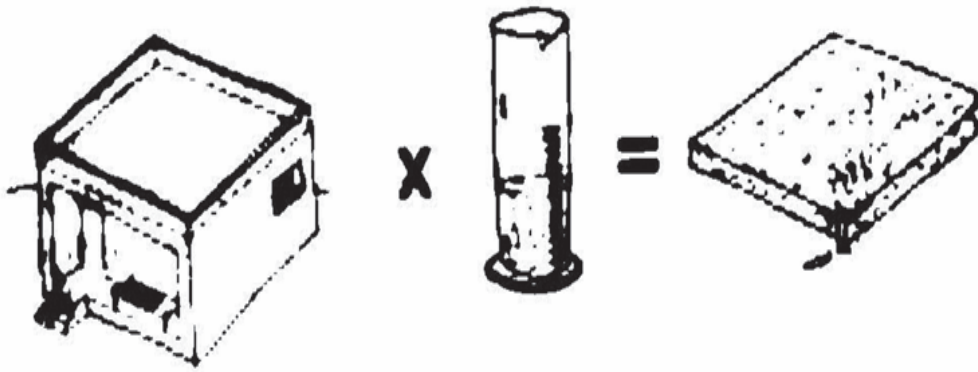
- Rainwater is bacteriological pure, free from organic matter and soft in nature
- Rainwater can be harnessed at place of need & time of need
- Recharged aquifers also serve as distribution system & water can be harnessed by just putting a hand pump at convenient location
- The infrastructure required for rainwater harvesting are simple & economical.

### **From Where to Harvest Rain Water:**

- Roof top to storage tank or Ground water recharging point
- Unpaved area direct seepage to ground water
- Water bodies like ponds, tanks and lakes
- Storm water drains directed to recharge ground water

## Rain Water Harvesting Potential

- The total amount of water that is received in the form of rainfall over an area is called the **rain water endowment** of that area.
- Out of this, the amount that can be effectively harvested is **called rain water harvesting potential**.



Area of catchment x Amount of rainfall = Rain water endowment



**Rain Gauge**

The collection efficiency is mainly dependent on factors like **runoff coefficient** and first flush wastage etc

## Runoff coefficients for various surface

| S. No. | Type of catchment Roof catchments                | Coefficients |
|--------|--------------------------------------------------|--------------|
| 1      | Tiles                                            | 0.8 - 0.9    |
| 2      | Corrugated metal sheets Ground surface coverings | 0.7 - 0.9    |
| 3      | Concrete                                         | 0.6 - 0.8    |
| 4      | Brick pavement Untreated ground catchments       | 0.5 - 0.6    |
| 5      | Soil on slopes less than 10%                     | 0.0 - 0.3    |
| 6      | Rocky natural catchments                         | 0.2 - 0.5    |

Runoff is the term applied to the water that flows away from catchments after falling on its surface in the form of rain

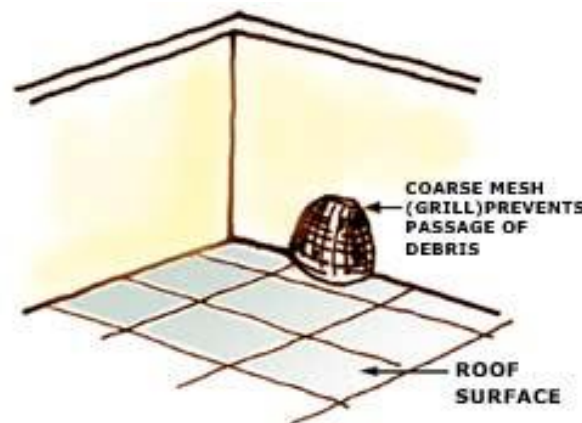
**Factors:** Rainfall pattern and quantity, catchment area characteristics etc. (10 years data required form metrological department)

# COMPONENTS OF A RAINWATER HARVESTING SYSTEM

## 1. Catchments:

- The catchment of a water harvesting system is the surface which directly receives the rainfall and provides water to the system.
- It can be a paved area like a terrace or courtyard of a building, or an unpaved area like a lawn or open ground.
- A roof made of reinforced cement concrete (RCC), galvanised iron or corrugated sheets can also be used for water harvesting.

## 2. Coarse mesh at the roof to prevent the passage of debris



### 3. Gutters:

Channels all around the edge of a sloping roof to collect and transport rainwater to the storage tank. Gutters can be semi-circular or rectangular and could be made using:

1. Locally available material such as plain galvanised iron sheet (20 to 22 gauge), folded to required shapes.
2. Semi-circular gutters of PVC material can be readily prepared by cutting those pipes into two equal semi-circular channels.
3. Bamboo or betel trunks cut vertically in half.



- The size of the gutter should be according to the flow during the highest intensity rain. It is advisable to make them 10 to 15 per cent oversize.
- Gutters need to be supported so they do not sag or fall off when loaded with water. The way in which gutters are fixed depends on the construction of the house; it is possible to fix iron or timber brackets into the walls, but for houses having wider eaves, some method of attachment to the rafters is necessary.

## 4.Conduits

Conduits are pipelines or drains that carry rainwater from the catchment or rooftop area to the harvesting system. Conduits can be of any material like polyvinyl chloride (PVC) or galvanized iron (GI), materials that are commonly available.

### Sizing of rainwater pipe for roof drainage

| Diameter<br>Of pipe<br>(mm) | Average rate of rainfall in mm/h |      |      |      |      |      |
|-----------------------------|----------------------------------|------|------|------|------|------|
|                             | 50                               | 75   | 100  | 125  | 150  | 200  |
| 50                          | 13.4                             | 8.9  | 6.6  | 5.3  | 4.4  | 3.3  |
| 65                          | 24.1                             | 16.0 | 12.0 | 9.6  | 8.0  | 6.0  |
| 75                          | 40.8                             | 27.0 | 20.4 | 16.3 | 13.6 | 10.2 |
| 100                         | 85.4                             | 57.0 | 42.7 | 34.2 | 28.5 | 21.3 |
| 125                         | -                                | -    | 80.5 | 64.3 | 53.5 | 40.0 |
| 150                         | -                                | -    | -    | -    | 83.6 | 62.7 |

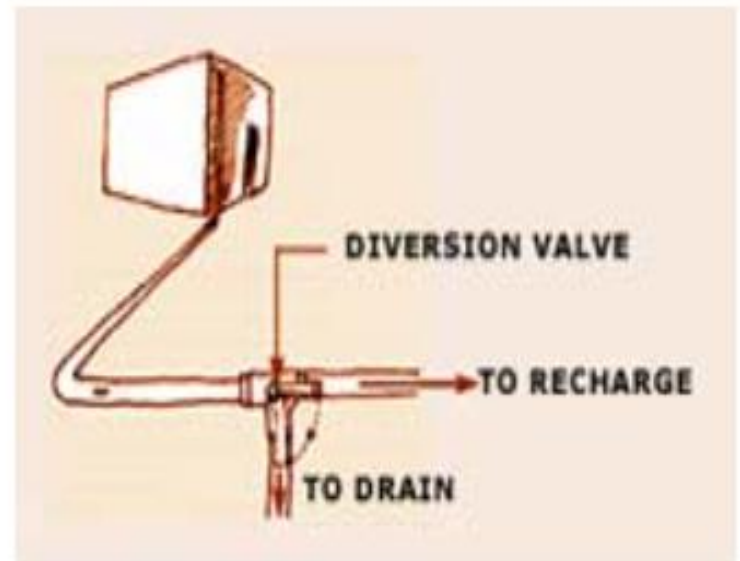
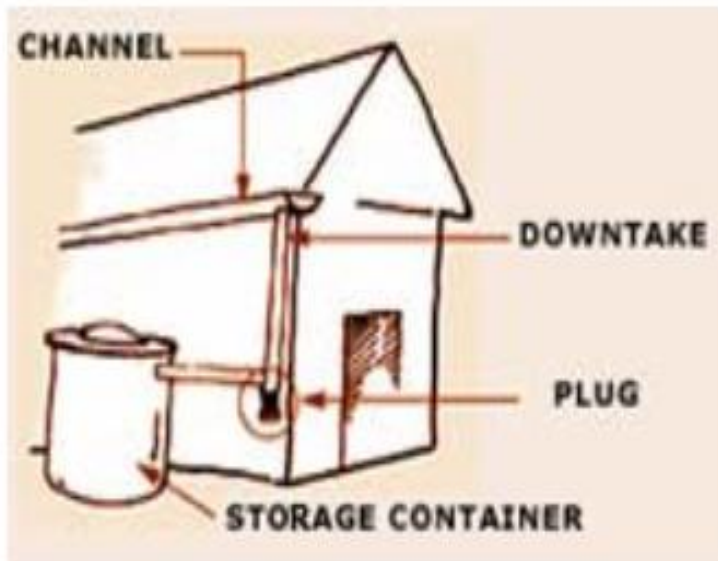
mm/ h - millimeters per hour; m - meters

Source: National Building Code

## 5.First-flushing

A first flush device is a valve that ensures that runoff from the first spell of rain is flushed out and does not enter the system.

- This needs to be done since the first spell of rain carries a relatively larger amount of pollutants from the air and catchment surface.



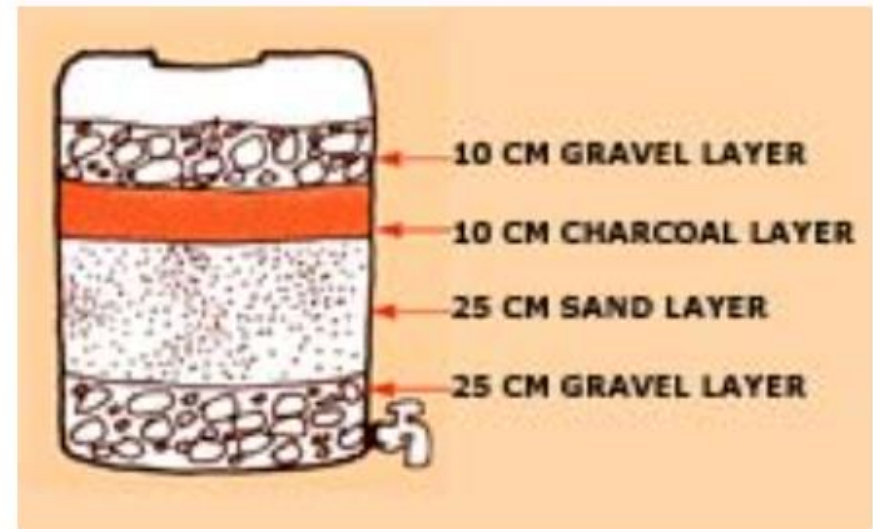
*Source: A water harvesting manual for urban areas*

## 6.Filter

The filter is used to remove suspended pollutants from rainwater collected over roof. A filter unit is a chamber filled with filtering media such as fibre, coarse sand and gravel layers to remove debris and dirt from water before it enters the storage tank or recharge structure. Charcoal can be added for additional filtration.

### (i) Charcoal water filter

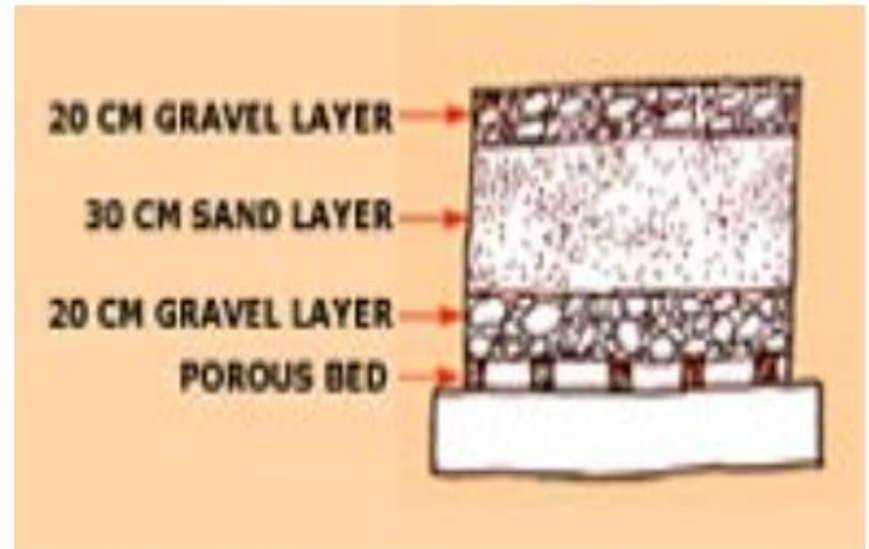
A simple charcoal filter can be made in a drum or an earthen pot. The filter is made of gravel, sand and charcoal, all of which are easily available.



*Source: A water harvesting manual  
for urban areas*

## **(ii) Sand filters**

Sand filters have commonly available sand as filter media. Sand filters are easy and inexpensive to construct. These filters can be employed for treatment of water to effectively remove turbidity (suspended particles like silt and clay), colour and microorganisms.



*Source: A water harvesting manual for urban areas*

## 7.Storagefacility

There are various options available for the construction of these tanks with respect to the shape, size and the material of construction.

**Shape:** Cylindrical, rectangular and square.

**Material of construction:** Reinforced cement concrete, (RCC), ferrocement, masonry, plastic (polyethylene) or metal (galvanised iron) sheets are commonly used.

**Position of tank:** Depending on space availability these tanks could be constructed above ground, partly underground or fully underground. Some maintenance measures like cleaning and disinfection are required to ensure the quality of water stored in the container.



A storage tank made of galvanised iron sheets

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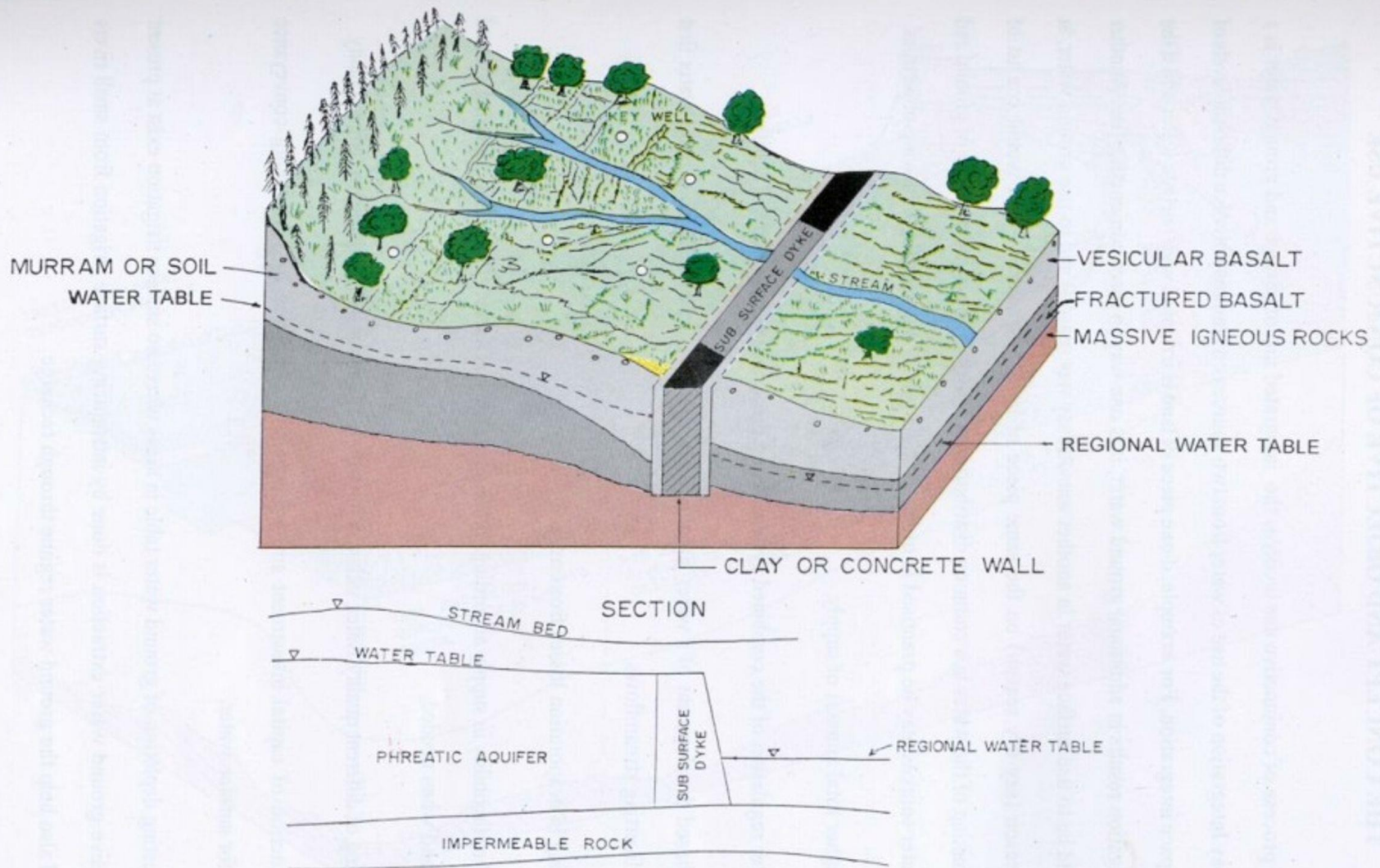
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# GROUND WATER DAMS OR SUB-SURFACE DYKES



## Subsurface barrier/Dykes

A **subsurface dyke** is a structure that is built in an aquifer with the intention of obstructing the natural flow of ground water, thereby raising the ground water level and increasing the amount of water stored in the aquifer.

- **Groundwater Management:** Helps in maintaining water levels in arid regions.
- **Flood Control:** Prevents excessive underground water movement that may lead to soil erosion.
- **Agricultural Benefits:** Enhances soil moisture, supporting sustainable farming.
- **Seawater Intrusion Prevention:** Protects freshwater aquifers in coastal areas.



# **Subsurface barrier/Dykes**

## **Types of Subsurface Dykes**

### **1.Geological Dykes (Natural)**

1. Formed by the intrusion of magma into pre-existing rock layers.
2. Often composed of igneous rock, such as basalt or dolerite.
3. Can act as groundwater barriers, influencing water flow in aquifers.

### **2.Artificial Subsurface Dykes**

1. Man-made structures built underground to control groundwater movement.
2. Used for water conservation, preventing seawater intrusion, or improving soil moisture retention in dry regions.
3. Typically made of clay, concrete, or impermeable membranes.

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## **Farm Ponding**

Farm ponds are small tanks or reservoirs constructed for the purpose of storing water essentially from surface runoff.

- Farm ponds are useful for irrigation, water supply for the cattle, fish production, etc.
- Floods can be controlled

### **Types of Ponds:**

Depending on the source of water and their location with respect to the land surface, farm ponds are grouped into four types.

- Dugout ponds or excavated ponds
- Surface ponds or watershed ponds
- Spring or creek fed ponds
- off-stream storage ponds



A **spring-fed pond** is primarily supplied by groundwater from a natural spring, while a **creek-fed pond** receives water from a flowing stream or creek

### **Design of Farm Ponds:**

The design of farm ponds consists of-

- (1) Selection of site,
- (2) Determination of the capacity of the pond,
- (3) Design of the embankment,
- (4) Design of the mechanical spillway,
- (5) Design of the emergency spillway, and
- (6) Providing for seepage control from the bottom.

## Contour trenching for rain water harvesting:

Contour trenching is a critical technique in sustainable land and water management.

Trenches are dug along contour lines across a slope to capture and retain water flowing downhill, allowing it to infiltrate the soil below.



### **Advantages:**

- Slowed-down water seeps into the ground, recharging the groundwater table.
- Prevents water from quickly running off the slopes, allowing more time for infiltration into the soil.
- Promotes sustainable agriculture on hill slopes.
- Contour trenches help prevent topsoil erosion, which is a major issue on slopes.
- Better water retention supports tree planting and afforestation efforts, which in turn help hold the soil and increase rainfall through transpiration.